AEGIS STATION

Master Dossier

v3.1 — 1g Configuration, Integrated Program Architecture · November 2026 · Supersedes v3.0

Aegis Station is a modular, shielded, rotating space habitat designed for construction in lunar orbit using attainable technology and lunar-derived water. It provides Earth-like living conditions through full artificial gravity and radiation protection, forming the backbone of a permanent human presence beyond Earth. The station is one element within a broader system-of-systems architecture that includes transportation, shielding logistics, autonomy, in-space assembly, and lunar surface operations.

---

1. Vision & Strategic Value

Aegis Station is designed from first principles around the realities of deep-space human habitation: radiation kills, zero-gravity degrades physiology, and resupply from Earth is prohibitively expensive. The program resolves all three constraints simultaneously — shielding from local lunar water, artificial gravity through rotation, and logistical self-sufficiency through in-situ resource utilization.

Full radiation shielding	3-meter water jacket sourced from lunar polar ice — no Earth-launched mass required
Artificial 1g gravity	Rotational spin at ~1.50 RPM; 1g at 397 m floor height
Massive usable volume	>1 million m³ per ring for residential, industrial, agricultural, and research use
Triple-ring redundancy	Three independent rings; any ring can be sealed without compromising station function
Cislunar logistics node	Integrated hub for lunar surface, orbital, and interplanetary operations
Replicability	Architecture scales to Mars orbit, Deimos, icy moons, or asteroid-belt outposts

---

2. Final Specifications

All figures reflect the 1g configuration with full shielding mass.

Ring centerline radius	350 meters
Tube (torus) radius	50 meters (full tube diameter: 100 m)
Outer hull radius	400 meters
Shield layer	47 – 50 m from tube center (3 m thick, lunar water)
Utility zone	40 – 47 m (pressurized; crew-accessible)

Pressurized hab zone	0 – 40 m inner radius within tube
Floor height	~397 m from rotation axis
Target gravity	~1.0 g at floor level
Spin rate	~1.50 RPM
Number of rings	3
Ring spacing	200 meters
Central hub length	650 meters (non-rotating)
Usable volume / ring	>1,000,000 m^3
Dry structure mass / ring	~220,000 metric tons
Shield mass / ring	~2,010,000 metric tons
Water-to-structure ratio	~9 : 1
Total shield volume	~6.0 million m^3 (~2,400 Olympic pools)
Total shield mass	~6.0 million metric tons

Shield volume is the shell between tube radii of 47 m and 50 m on the 350 m centerline: $2\pi^2 \times 350 \times (50^2 - 47^2) \approx 2.01$ million m^3 per ring, 6.03 million m^3 for three rings. The spin rate is the rate that gives 1 g at the 397 m floor.

---

3. Shield Architecture

Water as both structural shielding and an active operational system.

The radiation shield uses lunar-sourced water in a 3-meter layer flush against the outer hull. This approach eliminates dependence on Earth-launched shielding mass while creating a dual-use system: the same water volume supports life support, heat dissipation, and biosecurity functions.

Shield Structure

Physical layout and compartmentalization.

- 5 radial zones per ring with vertical segmentation
- Slosh suppression and resilience through compartment isolation
- Fill interfaces at 5 rotating vestibules per ring; top-offs through the hub/ring boundary

Fluids · CFD · Radiation · Compartmentalization

Active Systems

Fluid management and monitoring.

- Active circulation supporting ECLSS and heat rejection
- Flow control, filtration, and sensor-rich dynamic monitoring
- Fault isolation and rerouting on segment failure

Thermal · ECLSS · Sensors · Fault isolation

4. Shield Fill Operations

82 dedicated tankers executing a ~4.5-year initial fill campaign.

Water is delivered as modular 45-ton sealed cartridges, handled externally without pressurized transfer, and drained into ring reservoirs via centripetal flow once synchronized with ring rotation. Initial fill occurs with rings stationary; ongoing top-offs arrive as sealed cartridges through the hub/ring boundary at operational spin rate.

Cartridge format	Modular 45-ton sealed water cartridges
Tanker fleet (fill)	82 dedicated lunar tankers @ 45 MT payload each
Daily throughput	~3,690 metric tons / day
Initial fill duration	~4.5 years for complete shield across all rings
Loading method	External cartridge insertion into 5 rotating vestibules per ring
Draining method	Spin-synchronized centripetal flow into ring reservoirs
Initial fill protocol	Rings stationary; full shield mass loaded prior to spin-up
Operational top-offs	Sealed cartridges carried through the hub/ring boundary on cargo pods and discharged inside the ring at operational spin rate; inertial balance maintained

5. Construction & Deployment

Two-phase approach: dry assembly in orbit, then shield fill and spin-up.

Phase 1 — Dry Assembly

Structure-first, no water mass.

- Launch and assemble rings in lunar orbit — dry, no shielding
- Install shield bladders, plumbing, and fluid routing
- Integrate core utilities, docking arms, and central spine
- Commission life support and power systems pre-fill

Orbital assembly · Robotics · Structures · Logistics

Phase 2 — Fill & Spin-Up

Shield loading and station activation.

- Deliver water via tanker cartridge campaign (~4.5 years)
- Fill all three rings before initiating spin-up sequence
- Bring station to full 1g at ~397 m floor height
- Activate life support, utilities, and transport systems

ISRU logistics · Spin-up · ECLSS · Commissioning

6. Central Hub

650-meter non-rotating spine connecting all rings and hosting zero-G operations.

The central hub is the program's logistics and science backbone. It hosts docking, crew transit, zero-G workspaces, and the foundational infrastructure for expanding station capabilities over time.

Core Functions

Transit, docking, and operations.

- Docking arms at both ends — keep ring area clear for tanker access
- Pressurized transit pods for crew/cargo movement to rings
- Manual egress corridors as shielded backup routes
- Zero-G workspaces: labs, foundries, hydroponics, fabrication

Docking · Transit · Zero-G ops · Human factors

Radial Modules — Gravity Simulation

Variable-G research capability.

- Pressurized habitats on radial booms scale gravity from 0 to 1g
- Long-duration physiology trials at Mars and lunar gravity levels
- Agricultural tests and equipment validation under fractional G
- Pre-deployment testing for planetary settlement hardware

Research · Physiology · Agriculture · Hardware dev

7. Habitat Interior & Zoning

Over 1 million m³ of usable volume per ring, organized across functional zones.

Residential blocks	Private quarters, communal areas, and social infrastructure
Hydroponics & food systems	Continuous food production integrated with water and atmosphere loops
Industrial decks	Manufacturing, fabrication, and maintenance operations
Parks & social spaces	Open-area green space, recreation, and crew wellbeing environments
Medical & emergency areas	Trauma care, quarantine, and emergency response facilities
Combustion policy	Open flame permitted only inside hardened fire shelter modules

8. Emergency Preparedness

Redundancy is designed in, not bolted on.

Isolation & Redundancy

Fail-safe ring and life support design.

- Any ring can be sealed without compromising station function
- Life support sized for 3× the operational crew load
- Compartmentalized shielding: local failures don't compromise total protection
- Emergency routing via transit pods and manual egress corridors

Safety · Redundancy · ECLSS · Isolation

Inspection & Monitoring

Proactive hull and system health management.

- AHID (Automated Hull Inspection Drone) fleet for continuous monitoring
- Micrometeoroid damage detection, trending, and repair support
- ESIC (Extravehicular Structural Inspection Capability) for crew-assisted survey
- Sensor-rich shield fluid monitoring for pressure, flow, and contamination

Inspection · Autonomy · ESIC · Damage trending

9. Logistics Chain & Support Hardware

Material handling is a first-class engineering domain in a program of this scale.

The station's logistics architecture spans lunar surface extraction, orbital tanker operations, crew transport, and interplanetary freight. Each vehicle class is purpose-designed and integrated into a coherent supply chain.

Lunar Tanker Fleet

Autonomous surface-to-orbit water transport.

- 82 tankers, 45 MT payload each, sustained cadence operations
- Autonomous ascent/descent cycles with reliability-focused design
- Post-fill repurposing: resupply, depot missions, and fuel delivery

Autonomy · Propulsion · Fleet ops · ISRU

Luna–Aegis Shuttle

Reusable crew/cargo ferry.

- Short-range reusable lander: 2–6 crew + cargo
- Precision landing, surface handling, and rapid reflight
- Docking automation, crew safety, and life-support-in-transit

Reusable lander · Human rating · Docking · Avionics

Earth–Aegis Long-Hauler

Interorbital freight and passenger transport.

- Modular passenger and freight integration, mission-tailorable
- Depot-based refueling; rendezvous and docking automation
- Propulsion blocks, thermal management, and maintainable configurations

Transport · Propulsion · Thermal · Docking

Aegis-Class Rover

Pressurized surface operations platform.

- Surface scouting, prospecting, and ISRU site support
- Autonomous navigation in polar regolith and low-light terrain
- Docking/handling interfaces with shuttles and surface depots

Mobility · Autonomy · ISRU support · Human factors

---

10. Program Architecture & Technical Domains

Aegis Station is a system-of-systems program. The orbital habitat is one element within a broader architecture. Centralized integration authority preserves architectural intent while allowing flexible, best-in-class implementation across specialist teams.

10.1 Orbital Habitat Systems

Rotating structures, shielding, life support, power, and long-duration operability.

Large-Scale Rotating Structures

Structural engineering at full scale under 1g loads.

- Ring fabrication, modular segmentation, and orbital assembly tolerances
- Hub–ring interfaces, load paths, and structural margins under spin
- Materials, fatigue life, micrometeoroid tolerance, and maintainability

Aerospace structures · FEA / loads · Mechanisms · Orbital assembly

Rotational Dynamics & Stability

Spin-up, resonance, and mass management.

- Spin-up sequencing, damping, and vibration / modal management
- Mass asymmetry handling, trim, and operational balance constraints
- Attitude control interactions with rotating masses

Dynamics · Controls · Vibration · GNC

Radiation Shielding & Fluid Systems

Water shielding as protection and operational fluid.

- Segmented water containment, slosh suppression, and fault isolation
- Fill / top-off logistics, flow control, monitoring, and filtration
- Thermal coupling and heat rejection integration

Fluids · CFD · Thermal · Radiation

ECLSS

Long-duration crew systems built around reliability and safety.

- Air revitalization, water recovery, and waste processing
- Biosecurity, contamination control, and sensor-rich monitoring
- Fire containment strategies compatible with artificial gravity

ECLSS · Reliability · Safety · Human factors

Power Generation & Distribution

High-availability architecture with fault containment.

- Solar array systems, storage, and power management electronics
- Rotating-to-fixed power transfer, distribution buses, and protection
- Load shedding, redundancy zoning, and maintainable layouts

Power · Energy storage · Redundancy · Fault protection

Habitat Ops & Safety Architecture

Operating rules and physical design for continuity.

- Compartmentalization, isolation strategy, and emergency routing
- Maintainability, spares strategy, and on-orbit servicing provisions
- Operations planning, telemetry, and autonomy support

Ops · Safety · Maintainability · Autonomy

10.2 Transportation & Logistics Systems

Moving mass — water, propellant, cargo, crew — between lunar surface, orbit, and Earth.

Lunar Tanker Fleet

Autonomous surface-to-orbit water transport at sustained cadence.

- Autonomous ascent/descent cycles, turnaround, and reliability engineering
- Cartridge handling, transfer interfaces, and docking/berthing systems
- Fleet operations, maintenance regimes, spares, and performance tracking

Autonomy · Propulsion · Fleet ops · Ground systems

Luna–Aegis Shuttle

Reusable crew/cargo ferry linking polar sites and station.

- Precision landing, surface handling, and rapid reflight considerations
- Docking automation, crew safety, and life-support-in-transit
- Interfaces to rovers, depots, and station logistics

Reusable lander · Docking · Human rating · Avionics

Earth–Aegis Long-Hauler

Interorbital transport and cargo movement at scale.

- Modular freight and passenger integration, mission tailoring
- Propulsion blocks, power/thermal, and maintainable configurations
- Rendezvous, docking, and depot-based refueling interfaces

Transport · Propulsion · Thermal · Docking

Orbital Logistics & Handling

Material handling as a first-class system.

- Docking/berthing, pressurized transfer, and cargo staging
- Cartridge insertion/removal and inertial balance management
- Inventory tracking, spares logistics, and warehousing in orbit

Material handling · Interfaces · Inventory · Ops

10.3 Lunar Surface Systems

Extraction, processing, storage, mobility, and reliability through polar thermal extremes.

ISRU & Water Extraction

Volatile extraction feeding orbital logistics.

- Excavation and feed handling for icy regolith
- Thermal processing, separation, storage, and transfer hardware
- Quality control, contamination control, and system health monitoring

ISRU · Cryo / thermal · Process engineering · Reliability

Surface Mobility & Logistics

Rovers supporting prospecting, transport, and ops.

- Pressurized mobility, suspension/traction, and maintainability
- Navigation and autonomy in regolith, low-light, polar terrain
- Docking/handling interfaces with shuttles and surface depots

Mobility · Autonomy · Thermal · Human factors

Surface Power & Thermal Survival

Power for sustained ops and recovery modes.

- Solar + storage sizing, power management, and redundancy
- Thermal control for polar extremes and survival provisions
- Charging, staging, and depot-based logistics support

Power · Thermal · Night survival · Depots

Surface Ops & Field Maintenance

Keeping systems running is part of the architecture.

- Field-replaceable modules, toolchains, and repair workflows
- Diagnostics, telemetry, and remote support pathways
- Spare parts logistics and reliability planning

Ops · Maintainability · Telemetry · Reliability

10.4 In-Space Assembly & Maintenance

Assembling, inspecting, and sustaining large infrastructure in lunar orbit over decades.

Modular Orbital Assembly

Robotic and human-compatible assembly of large structures.

- Assembly robotics, manipulators, fixtures, and alignment methods
- EVA tool compatibility and maintainable interface design
- Construction sequencing, tolerance stack-up, and verification

Robotics · EVA · Construction · Verification

ESIC — Extravehicular Structural Inspection

External inspection as an operational capability.

- Hull inspection, micrometeoroid damage detection, and trending
- Autonomous and crew-assisted inspection workflows
- Repair support provisions and interface to maintenance planning

Inspection · Autonomy · Repair support · Ops

10.5 Program-Level Integration

Preserving architectural intent while enabling flexible implementation across organizations.

Systems Architecture Authority

Centralized authority for intent, performance, and trades.

- Architecture baselines, requirements flowdown, and trade studies
- Design intent preservation and end-state alignment
- Cross-domain coordination and decision discipline

Architecture · Trade studies · Requirements

Interface Control & Configuration Management

Where large programs succeed or fail.

- Interface Control Documents (ICDs) and verification ownership
- Configuration baselines, change control, and traceability
- Integration sequencing and readiness gates

ICDs · Config control · Verification

Integration, Test & Verification

Demonstrating the system meets intent at program scale.

- Incremental integration plans and system-level test strategy
- Simulation, hardware-in-the-loop, and operational rehearsal
- Independent verification and validation where appropriate

IT&V · Simulation · Readiness

Program Management & MBSE

Schedule, risk, cost, and model-based traceability.

- Risk management, schedule discipline, and milestone governance
- Model-Based Systems Engineering (MBSE) for traceability and integration
- Operations planning and lifecycle sustainment considerations

PM · Risk · MBSE · Lifecycle

Execution Model

Best-in-class subsystem implementation paired with centralized integration authority. Implementation remains flexible. Architectural intent does not. Domain teams with relevant heritage — space-rated systems, lunar surface operations, high-reliability autonomy, large-scale structures, ECLSS, ISRU, or orbital assembly — are designed to integrate as specialist contributors into a coherent, buildable program.

Open architecture · Specialist integration · Heritage welcome

11. Open Engineering Topics

Active development areas requiring further study and experimental work.

Slosh dynamics	Rotational fill under spin; multi-compartment fluid behavior at habitat scale
Segment fault handling	Fluid rerouting, pressure equalization, and isolation on compartment failure
Coriolis effects	Large-volume water flow management under continuous rotation
Microbial suppression	Long-term biosecurity inside the shield fluid over multi-decade operational life
Mass asymmetry	Structural and dynamic implications during partial-fill stages and asymmetric damage
Spin-up resonance	Modal response under combined structural and fluid mass during spin-up sequencing
Shield replenishment rate	Minimum viable top-off cadence balancing shield integrity with tanker fleet sizing

12. Replicability & Growth

The Aegis architecture is not a one-off. Any location with accessible local water — Mars orbit, Deimos, icy moons, or asteroid-belt outposts — becomes a viable site for an Aegis-type installation. The program's modular design, standardized interfaces, and logistics framework are intentionally transferable, forming the repeatable backbone of a multi-node interplanetary civilization.

Lunar orbit	Primary program location; water from polar ice deposits
Mars orbit	Phobos/Deimos water access; cislunar logistics patterns replicate directly
Icy moons	Europa, Ganymede, Callisto as long-range staging grounds
Asteroid belt	Compatible with water-rich C-type asteroid mining operations
Expansion	Modular add-ons and docking provisions for mission-specific growth at any node

13. Changes From v3.0

The geometry is unchanged. The figures derived from it are recomputed, and the structure is resized.

Item	v3.0	v3.1
Spin rate	~1.33 RPM	~1.50 RPM, the rate that gives 1 g at the 397 m floor
Shield mass / ring	~1,100,000 metric tons	~2,010,000 metric tons, the full 3 m shell between tube radii of 47 m and 50 m
Total shield volume	~3.3 million m³ (~1,320 Olympic pools)	~6.0 million m³ (~2,400 Olympic pools)
Total shield mass	~3.3 million metric tons	~6.0 million metric tons
Dry structure mass / ring	~120,000 metric tons	~220,000 metric tons, resized for the spin and pressure loads with conventional materials
Water-to-structure ratio	~27 : 1	~9 : 1, shield mass per ring over dry structure mass per ring
Tanker fleet (fill)	45 tankers	82 tankers, sized to hold the ~4.5-year fill
Daily throughput	~2,025 metric tons / day	~3,690 metric tons / day
Operational top-offs	Cartridges loaded at the rotating vestibules at spin rate	Sealed cartridges carried through the hub/ring boundary on cargo pods